

Project Outline: Glasma Role in Jet Quenching Effects

① Scientific Context (*INTRODUCTION, STATE OF THE ART*)

- Heavy Ion Collisions (HICs), Jet Quenching.
- Initial Stage of HICs (Glasma).
- Color Glass Condensate (CGC).
- Glasma role in JQ: current state of knowledge.

Suggested reading: [arXiv:1802.04801](#), [arXiv:0712.3443](#), [arXiv:1401.4866](#), [arXiv:2112.06812](#) (and references therein)

② Groundwork: Calculation of Pre-equilibrium Potentials in the McLerran-Venugopalan (MV) Framework

(a) Solving Yang-Mills (YM) equations for a single nucleus

→ First in covariant gauge, then transforming to light-cone gauge.

[arXiv:hep-ph/0011241](#), [arXiv:hep-ph/0402256](#)

(b) Calculating Glasma fields

→ Propose ansatz (in Fock-Schwinger gauge) and use it to compute fields at $\tau=0^+$.

→ To do that, one needs to re-write the (homogeneous) YM equations in terms of the aforementioned ansatz (and in the Bjorken coordinate system) and require them to be regular at $\tau=0^+$ (by replacing the ansatz, taking $x^\pm \rightarrow 0$ limit, and canceling singularities).

(c) Writing chromo-electric and chromo-magnetic fields at $\tau=0^+$

→ Express in terms of pre-collision potentials.

[arXiv:hep-ph/9502289](#), [arXiv:hep-ph/0602189](#)

③ Medium Averages

- Understanding the physical picture (from both CGC and JQ perspectives) *[arXiv:0712.3443](#)*
- Understanding Glasma Graph approximation
- Specific objects to compute (for $\tau=0^+$):

(a) Two-point function of pre-equilibrium gauge fields

→ Using two approaches: GBW and MV models

→ Understanding the differences between them

→ Running coupling

(b) Writing $\frac{dE}{dx}$ and $\hat{q}$ in terms of the two-point function

→ Obtain explicit expressions in GBW and MV models

[arXiv:1808.00795](#), [arXiv:2102.09993](#), [arXiv:2202.00357](#)

④ Proper Time Evolution

- Understanding our approach (linearizing YM equations) and differences with other strategies (e.g. numerical approaches, expansion of gluon fields in powers of τ)

(a) Solving linearized YM equations

→ Using Coulomb gauge

→ Matching resulting fields to initial conditions at $\tau=0^+$

→ Understanding why this is an appropriate approximation

(b) Obtaining τ -dependent expressions

[arXiv:2102.09993](#), [arXiv:1708.08625](#), [arXiv:2101.03791](#), [arXiv:2202.00357](#)

⑤ Evaluation of Expressions: Obtaining curves for dE/dx and $\hat{q}$

- Numerically (analytically if possible, or under certain approximations)
- Evaluate dependence on IR cutoff and Q_s

[arXiv:2202.00357](#)

⑥ Assessment of Impact of Glasma Phase on JQ (i.e. transverse momentum broadening)

(a) Numerical estimation

(b) Comparison to broadening during hydro phase

[arXiv:2202.00357](#)

⑦ Conclusions and Future Prospects